Computationally Sufficient Memory: Causal Native-K/V Materialization for Progressive Retrieval Attention

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August 28, 2026

Abstract

Sparse memory systems often collapse two operations: finding a relevant region and activating all of its key/value (K/V) states. We separate them and ask which already-contextualized native states are computationally sufficient once conceptual selection is fixed. Progressive Retrieval Attention (PRA) materializes source-relative logical intervals across storage shards while preserving resource boundaries, token order, deduplication, and grouped-query-attention shapes.

The primary experiment reuses 25 frozen controlled-model checkpoints: five seeds at local receptive fields $W \in \{16, 32, 64, 128, \text{global}\}$. For each checkpoint, oracle parent identity is fixed while exact evidence cores, local radii, complete facts, whole parents, matched wrong memory, fixed budgets, and gist controls vary. Across 200 held-out checkpoint–example pairs, exact cores use 5.25 K/V tokens on average, reduce K/V by 92.9% relative to whole parents, raise the correct-answer margin by 4.107 over no memory, and increase accuracy from .080 to .720. Whole-parent disclosure raises margin by only .981 and reaches .300 accuracy. Matched wrong cores use the same K/V but lower margin by .798, establishing a content-causal effect. Broader radii monotonically lower evidence attention and output quality; the result is stable across all five receptive-field settings. Consumer layer matters, while contextual representation change is measurable but does not predict the exact-core advantage.

We then confirm the mechanism with frozen Qwen3-0.6B using 16 validation and 32 held-out examples per dataset. Exact evidence reduces active K/V by 20.9% on MuSiQue and 44.4% on 2WikiMultiHopQA while preserving the corresponding whole-parent result within paired uncertainty. A transport audit subsequently finds that the inherited replay restarted direct-query RoPE positions at zero. Under corrected source-offset transport and a separate 16-example-per-dataset layer calibration, contiguous late consumers dominate scattered placement. The last 20 of 28 layers improve gold-answer log probability over no memory by 1.946 nats on MuSiQue and 1.276 on 2Wiki; the last eight retain 1.190 and .949 at 40% of that layer-token cost. All-layer replay is not a quality ceiling: it reaches .957 and -2.290, respectively, because additional memory consumers can amplify interference. Native gist means do not substitute for token detail. These experiments support a precise distinction: annotation evidence identifies justificatory spans, whereas computationally sufficient memory specifies the contextual native states required by a particular consumer. The paper establishes a memory-quality frontier, not a production speed claim.

1 Introduction

A retrieval system answers at least three questions. Which memory is relevant? Which physical detail should become active? Can the model use that detail to produce the answer? These questions are easy to collapse because ordinary retrieval-augmented generation serializes selected text into one prompt. Once a document is selected, all of its re-tokenized prompt state participates in attention. Native-K/V memory exposes the separation more directly. A compact representation can identify a conceptual region while the attention kernel receives a smaller set of already computed K/V states.

PRA Paper 1 separated logical memory from active attention state [5]. Paper 1.5 showed why semantic routing state and native positional attention state should remain distinct [4]. Paper 2 integrated that boundary into pretrained Hugging Face decoder families [6]. Paper 2.5 then treated discovery as bounded traversal over conceptual memory regions. The present paper asks the next question:

> Once relevant conceptual memory is known, how little native K/V detail must be materialized for a frozen model to preserve useful output behavior?

The primary experiment begins with a controlled oracle. A generated chain fixes one conceptual parent and marks the exact tokens that carry its facts. A router therefore cannot take credit or blame for the result. We vary exact cores, local context, complete facts, whole parents, fixed budgets, native gist means, and matched wrong memory. Only after diagnosing this controlled frontier do we return to dataset annotations in a frozen Qwen model and to Paper-2.5's learned selectors. The experimental identity is therefore

$$Q_{\text{output}} = f(M \mid S), \tag{1}$$

where selection S is fixed within a comparison and materialization M alone changes.

This distinction matters economically and scientifically. Broad conceptual search can improve recall, but whole-parent disclosure makes breadth expensive in K/V and introduces irrelevant softmax competitors. Conversely, a tiny physical budget can be evidence-dense while omitting required evidence regions. We therefore report both evidence density and evidence coverage, rather than treating either as sufficient.

The paper makes five contributions:

1. It defines *computational sufficiency* as a physical-memory question distinct from retrieval relevance and from the consumer's learned ability to use memory.

2. It provides a model-independent logical interval abstraction and exact cross-shard native-K/V gather with deduplication, resource isolation, GQA preservation, and transfer accounting.

3. It introduces a controlled oracle intervention that matches physical K/V while changing content, measures attention and residual trajectories, and varies receptive field, consumer layer, representation context, evidence dispersion, and physical budget.

4. It tests whether the controlled diagnosis transfers to frozen Qwen3-0.6B on MuSiQue and 2Wiki with validation-frozen policies and paired example-level confidence intervals.

5. It audits positional replay and calibrates contiguous and sparse consumer-layer profiles under corrected native-K/V transport, separating address, detail-storage, routing, and consumption roles.

6. It preserves negative results: broader disclosure can dilute useful evidence, gist means are not token-detail substitutes, high density can hide low coverage, and Python gathering does not yet establish a serving-speed benefit.

The next section separates relevance from physical sufficiency. We then describe the interval materializer, use the controlled model as a causal microscope, and finally test which observations survive in a pretrained language model.

2 Computationally Sufficient Memory

2.1 Selection, materialization, and use

Let $\mathcal{C}$ denote conceptual memory regions, $S(q, \mathcal{C})$ a selection procedure, $M(S, B)$ a materializer under physical budget B, and U the frozen transformer's native attention and generation path. PRA computes

$$C_q = S(q, \mathcal{C}), \tag{2}$$
$$(K_M, V_M) = M(C_q, B), \tag{3}$$
$$y = U(q, K_M, V_M). \tag{4}$$

Paper 2.5 studies the first line. This paper freezes the first line and studies the second. A memory-use adapter would change the third line and is outside the primary experiment.

This factorization separates two meanings of "evidence." An annotation answers a semantic question: which source spans justify an answer to a human reader? A materializer answers a computational question: which contextual native K/V states must remain active for this model, at this consumer layer, to preserve useful computation? The two sets can coincide, but no definition requires them to do so. Nearby tokens may supply entity resolution or syntax; alternatively, they may only add softmax competitors. We call a disclosure computationally sufficient when it reaches a predeclared fraction of a physical control's output gain under fixed selection and model weights.

This factorization prevents two misleading conclusions. First, high evidence recall does not prove that whole selected parents help output; their irrelevant K/V can dilute attention. Second, poor output after a small materialization does not prove that selection failed; the selected concept may be correct while its disclosed detail lacks context.

Whole-parent memory is therefore a local physical control, not an oracle optimum. If an exact core outperforms its parent, the correct interpretation is distractor dilution, not retention above an assumed ceiling. If the parent itself fails to improve no memory, percentage retention is left undefined rather than converted into a misleading ratio.

2.2 Three granularities

We record three independent granularities:

$$G_{\text{encode}} \neq G_{\text{search}} \neq G_{\text{materialize}}. \tag{5}$$

Canonical contextual K/V is encoded in model-safe blocks of 256 tokens. Search addresses 32-token parents. Materialization operates on arbitrary source-relative token intervals. Storage boundaries therefore do not define semantic windows.

2.3 Logical materialization domains

An interval is a half-open tuple

$$I = (d, s, e), \qquad 0 \leq s < e, \tag{6}$$

where d is a logical domain. Every explicit URI is a separate domain. A window near the end of one resource cannot continue into another resource merely because their tensors are adjacent in a cache. The displaced long-prompt resource `#__head` and the live prompt tail are one exception: they share the continuous prompt-history domain established by canonical PRA.

For annotated evidence $[s_i, e_i)$, a symmetric local policy requests

$$I_i(r) = [\max(0, s_i - r), \min(L_d, e_i + r)), \tag{7}$$

with clipping inside domain d. The physical request is the domain-wise union

$$\mathcal{I} = \bigcup_i I_i, \tag{8}$$

so each logical token is materialized at most once.

3 Materialization Mechanism

3.1 Stored shards and canonical provenance

Each reference layer stores shards with logical span, source URI, canonical post-position K/V, and encoding provenance. Primary experiments never re-encode isolated evidence. If a semantic window crosses two routing chunks, the gatherer slices both stored shards and concatenates them in logical order. For grouped-query attention, K/V remain

$$[1, H_{kv}, T, D_h], \tag{9}$$

without expansion to query-head count.

Figure 1 shows the separation. Conceptual selection stops before the physical materializer. The direct prompt K/V and selected memory K/V meet only at native attention.

logical source / long prompt → model-safe contextual encoding blocks → canonical native K/V shards

↓

smaller routing parents → gists / packed conceptual index → fixed oracle or learned selection

↓

logical interval planner → domain union and deduplication → cross-shard gather under K/V budget

↓

selected native K/V + direct prompt K/V → pretrained native attention

Figure 1: Selection and physical disclosure are separate operations. Search parents are not required to be encoding blocks or materialization windows.

3.2 Cross-shard gather

For each merged interval, the resolver advances a logical cursor. Among overlapping storage shards covering the cursor, it selects the shard extending furthest right, slices only the uncovered range, and repeats until the interval ends. Missing coverage raises an error instead of borrowing tokens from another URI. CPU-resident slices are collected before transfer; only the selected slices move to the GPU.

Listing 1: Simplified logical K/V gathering.

```
merged = union_intervals(requested_intervals)
```

```
for interval in merged:
    cursor = interval.start
    while cursor < interval.end:
        shard = furthest_shard_covering(interval.domain, cursor)
        stop = min(interval.end, shard.end)
        k_parts.append(shard.k[:, :, cursor-shard.start:stop-shard.start, :])
        v_parts.append(shard.v[:, :, cursor-shard.start:stop-shard.start, :])
        cursor = stop
K = torch.cat([part.to(device) for part in k_parts], dim=2)
V = torch.cat([part.to(device) for part in v_parts], dim=2)
```

The implementation records requested tokens before union, deduplicated tokens, materialized bytes, evidence and non-evidence tokens, shards touched, cross-shard interval count, interval-resolution time, gather time, and host-to-device time.

3.3 Fixed budgets across evidence regions

For n regions with capacities c_i, a fixed budget assigns integer quotas b_i satisfying

$$1 \leq b_i \leq c_i, \qquad \sum_i b_i \leq B. \tag{10}$$

Equal, evidence-length-proportional, minimum-core-plus-remainder, and score-proportional allocation are implemented. The preserved pretrained pilot uses equal allocation. Every region receives at least one token, but a budget smaller than the complete evidence cores may explicitly truncate those cores. Evidence coverage exposes that truncation; it is never inferred from density.

4 Controlled Disclosure Ladder

The controlled model gives every policy the same query, weights, oracle parent, and consumer layer. Only the native K/V positions made visible to PRA change. Table 1 begins with no memory, adds progressively broader evidence-centered detail, and ends with controls that separate content from budget and gists from token states.

Table 1: Controlled toy materialization policies. A fact contains five serialized tokens; its three-token semantic core excludes framing tokens.

Policy	Name	Physical disclosure
T0	no memory	Direct query only
T1	exact core	Union of the three-token evidence cores
T2–T5	local radii	Core plus $r \in \{2, 4, 8, 16\}$ tokens on each side
T6	complete facts	Complete five-token serialized evidence facts
T7	whole parent	Every native K/V state in the selected source block
T8	fixed budgets	12, 24, or 48 K/V tokens under three region-allocation rules
T9	wrong exact	Equal-budget semantic cores from matched distractor facts
T10	native gist	One mean native K/V state for the selected parent
T11	gist plus exact	Native gist together with the T1 token detail

T9 is the key causal control. T1 and T9 expose the same number of native states at the same layer, but only T1 carries the answer chain. A margin difference between them therefore cannot be explained by a larger memory tensor. T7 answers a different question: whether retaining every parent token helps the frozen consumer. It is not assumed to be the best condition.

The native gist in T10/T11 is deliberately not the routing vector. On a GQA model, routing width can equal hidden size while native K/V width is $H_{kv}D_h$. We average the parent's actual native K and V independently over tokens, preserving physical head layout. The pretrained confirmation uses analogous M-series conditions: no memory, gist, whole selected parents, exact annotated evidence, and radii 2, 4, and 8 on validation. Held-out evaluation retains exact evidence, radius 2, and the validation-selected radius when distinct.

5 Controlled Experimental Protocol

The controlled task is deliberately small enough to support intervention rather than correlation. Each source serializes a randomized chain of one to four facts among distractors. The query asks for one of eight balanced labels, and the generator randomizes entities, relations, lexical overlap, branch structure, evidence gaps, and distractor count. The evidence core and every distractor core are known exactly. Sixteen deterministic examples are divided into eight validation and eight held-out identities.

We reuse the final Paper-2.5 checkpoints rather than retraining for this paper. Five seeds are available for each receptive field $W \in \{16, 32, 64, 128, \text{global}\}$, giving 25 frozen models. All policies share the same layer-native K/V computed from one contextual encoding of the complete source. The primary consumer is layer 0; layers 2 and 5 form a predeclared consumer-layer profile. Representation portability is measured at layer 2 by comparing aligned value states from the contextual parent and from isolated evidence encoding. Table 2 records the full design.

Table 2: Controlled causal materialization protocol.

Property	Value
Task	Eight-label randomized relational chains, depth 1–4, with matched distractors
Frozen models	25 checkpoints: 5 seeds × 5 receptive fields
Validation / held-out	8 / 8 identity-disjoint generated examples per checkpoint
Selection	One oracle parent; parent identity cannot vary across conditions
Primary / profile layers	Layer 0 primary; layers 0, 2, and 5 profiled
Policies	T0–T11, radii 0/2/4/8/16, budgets 12/24/48
Primary metric	Correct-answer margin change relative to T0
Additional metrics	Accuracy, probability, NLL, Brier score, attention partitions, residual trajectory, K/V, latency
Hardware	NVIDIA GeForce GTX 950M, CUDA; weights frozen for every intervention

The primary output is the margin between the correct label and the strongest wrong label. We also record correct-class probability, NLL, entropy, Brier score, and accuracy. Final-query memory attention is partitioned into evidence, surrounding non-evidence, and distractor mass; the remaining softmax mass belongs to the direct query. Residual hooks recover the intermediate answer margin after every layer, permitting immediate-gain and later-erasure measurements.

6 Controlled Results

6.1 Exact evidence is causally useful

The controlled intervention resolves the first question before any pretrained benchmark is examined. Across five receptive fields, five seeds, and eight held-out examples per checkpoint, T1 activates 5.25 native K/V tokens on average. It raises mean answer margin from −2.694 to 1.413 and accuracy from .080 to .720. T9 activates the same number of states but lowers margin to −3.492 and reaches .050 accuracy. The T1–T9 margin gap is therefore 4.905 points under a matched physical budget.

Whole-parent disclosure averages 73.88 K/V tokens, a 14.1-fold larger payload. It raises margin by only .981 over no memory and reaches .300 accuracy. Exact cores improve margin by 4.107 and reduce K/V by 92.9% relative to that parent. Complete five-token facts sit between the two: 8.75 K/V tokens, a 3.674-point margin gain, and .635 accuracy. The model needs the evidence content, but the framing tokens are not helpful on average.

Table 3: Controlled held-out frontier. Means cover five windows, five seeds, and eight held-out examples per checkpoint (200 checkpoint–example pairs per row). Δm is margin change from no memory.

Disclosure	K/V	K/V reduction	Δm	Accuracy	Evidence mass
No memory	0.00	–	.000	.080	.000
Wrong exact	5.25	92.9%	−.798	.050	.000
Native gist	1.00	98.6%	.238	.110	.000
Exact core	5.25	92.9%	4.107	.720	.604
Complete fact	8.75	88.2%	3.674	.635	.527
Radius 2	12.25	83.4%	3.340	.570	.466
Radius 4	18.50	75.0%	3.111	.560	.389
Radius 8	27.25	63.1%	2.190	.410	.292
Radius 16	41.38	44.0%	1.438	.340	.215
Whole parent	73.88	0.0%	.981	.300	.149

Figure 2 makes the physical frontier visible. The strongest point is not the largest disclosure. Once the oracle core is active, each ring of surrounding K/V lowers both its share of attention and the final margin.

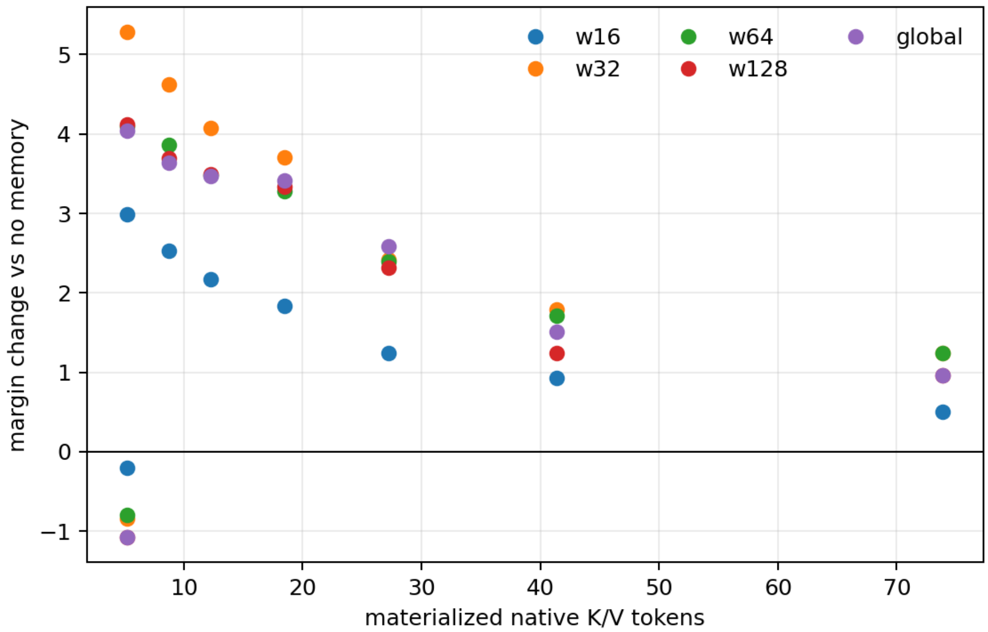

Figure 2: Held-out margin change against materialized native K/V. Color denotes the checkpoint's training receptive field. Exact-core points occupy the upper-left frontier.

6.2 Receptive field does not reverse the result

The $W\times$materialization sweep tests whether a memory state trained with broader native context requires a broader physical neighborhood when replayed. Table 4 rejects that candidate signature in this task. Exact cores outperform whole parents at every W, including the global model. The advantage is largest at $W = 32$, but there is no monotonic increase in the required radius. Validation selects radius zero for $W = 16, 32, 64$. For $W = 128$ and global, the whole parent itself lowers validation margin relative to no memory, so percentage retention is undefined rather than reported as a radius.

Table 4: Held-out receptive-field interaction. Each row averages five seeds and eight examples.

W	Exact Δm	Exact acc.	Parent Δm	Parent acc.	Exact–parent margin
16	2.994	.625	.501	.175	2.493
32	5.281	.825	1.238	.250	4.043
64	4.094	.725	1.236	.350	2.858
128	4.121	.750	.964	.375	3.157
global	4.047	.675	.963	.350	3.084

The validation-selected sufficiency calculation in Table 5 uses whole-parent margin gain only when that gain is positive. Exact cores exceed 95% of the gain for the three local-W models. The two broader models are marked unresolved because their validation parent control is worse than no memory; assigning them radius zero would hide that failed denominator.

Table 5: Validation-selected disclosure for 95% of whole-parent margin gain. Portability is mean layer-2 $1 - \cos$ change between contextual and isolated evidence values.

W	Selected radius	Held-out K/V	Reduction vs parent	Portability
16	0	5.25	92.9%	.286
32	0	5.25	92.9%	.270
64	0	5.25	92.9%	.218
128	unresolved	–	–	.157
global	unresolved	–	–	.143

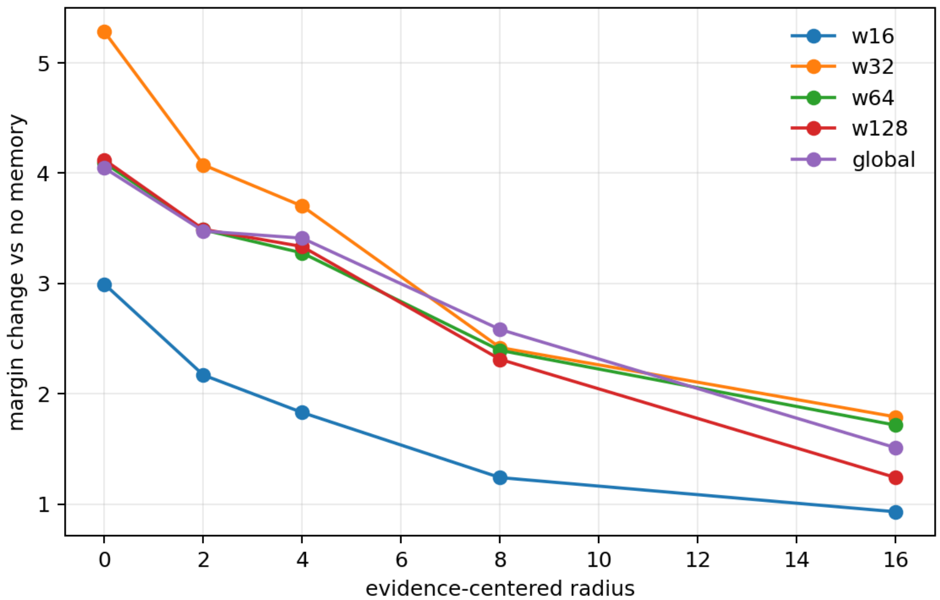

Figure 3: Evidence-centered radius versus held-out margin gain. Broader disclosure lowers quality for every receptive-field setting; global attention does not require a broader replay window.

6.3 Attention dilution and residual persistence

Evidence attention falls from .604 at the exact core to .466 at radius 2, .292 at radius 8, and .149 for the parent. The direct-query share also falls as memory grows. The added states are therefore not free background: they consume the same normalized attention budget. Figure 4 pairs this decomposition with the residual trajectory. Useful disclosure changes the answer geometry immediately and much of that displacement survives later layers; wrong memory moves the margin in the opposite direction.

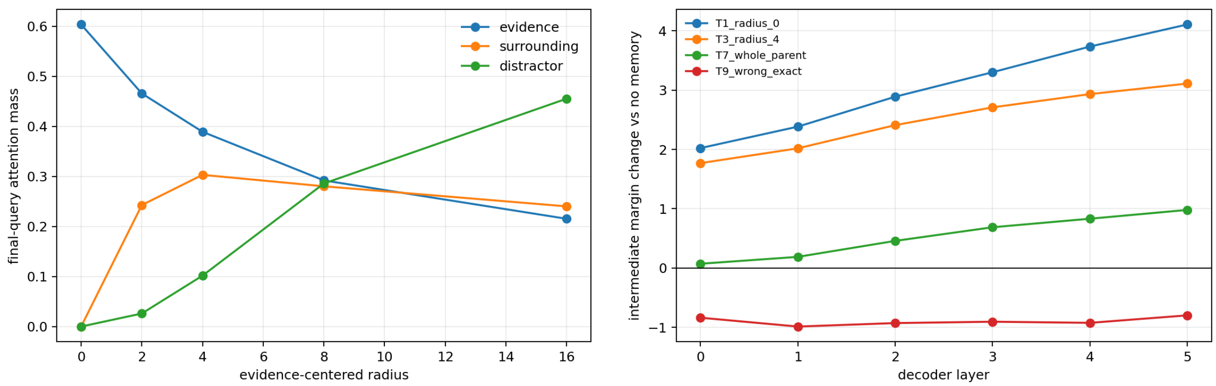

Figure 4: Left: final-query attention partitions as radius grows. Right: answer-margin trajectory after disclosure at the primary consumer.

Consumer placement changes the final effect. Exact-core injection at layers 0, 2, and 5 yields mean margin gains of 4.107, 2.369, and .554. The attention mass remains substantial at all three, so attention alone does not imply that later residual computation can exploit the update. This 3.554-point range supports a consumer-layer mismatch diagnosis, but the pretrained confirmation preserves inherited Paper-2.5 layer bands to avoid tuning Qwen on the controlled outcome.

6.4 Portability, dispersion, and budgets

Aligned layer-2 values change by mean cosine distance .215 when the same evidence is encoded in its parent instead of isolation. Representation context dependence is therefore real. Its correlation with the exact-core minus parent margin is only $r = .026$, however, so it does not explain the observed advantage. Figure 5 records this null relationship rather than promoting context sensitivity into a causal claim.

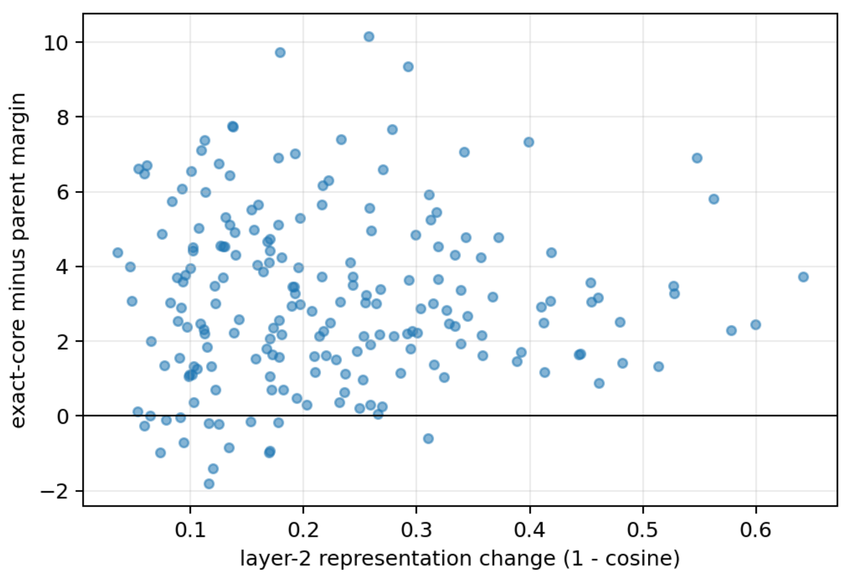

Figure 5: Layer-2 representation change versus exact-core advantage. Context changes native value states, but variation in that change does not predict disclosure quality in this controlled task.

At fixed 12 evidence tokens, replacing one contiguous envelope with two disjoint regions reduces active K/V from 121 to 44 and changes margin relative to no memory from −.592 to +1.008; four regions produce the same union in this generator. This is a geometry result, not evidence that more reasoning hops are inherently easier. It shows why a materializer must preserve disjoint logical regions instead of filling the gaps between them.

Table 6: Region dispersion at fixed evidence amount. Grouping one region fills the intervening source envelope; two and four groups preserve the same disjoint interval union in this corpus.

Physical regions	Evidence tokens	Active K/V	Margin change vs no memory
1	12	121	−.592
2	12	44	+1.008
4	12	44	+1.008

Budgets 12, 24, and 48 all preserve complete cores, yet their margin gains decrease from 3.146 to 2.345 and 1.665 as non-evidence states accumulate. Equal, evidence-proportional, and minimum-core-plus-remainder allocation coincide because every controlled core has equal length; the allocation comparison is therefore unresolved. The diagnosis freezes exact cores and radius 2 for confirmation, preserves complete cores before remainder, and leaves adaptive allocation to a task with unequal evidence regions.

Table 7 freezes the causal diagnosis before pretrained confirmation. The ranking distinguishes an observed representation property from a demonstrated output failure.

Table 7: Controlled causal diagnosis on held-out rows.

Candidate failure mode	Status	Evidence
Distractor dilution	supported	Exact core exceeds parent margin by 3.127 and matched wrong core by 4.905.
Consumer-layer mismatch	supported	Exact-core effect spans 3.554 margin points across layers 0, 2, and 5.
Representation-context dependence	partial	Mean value-state change is .215, but correlation with disclosure advantage is .026.
Missing contextual support	unsupported	Radius 2 lowers margin by .768 relative to exact core.
Evidence incompleteness	unsupported	Exact and radius policies retain complete semantic-core coverage.

7 Pretrained Qwen Confirmation

7.1 Protocol

The controlled study predicts that exact evidence should be a strong physical point and that broader local context may dilute it. We first tested that prediction without changing Qwen3-0.6B weights, conceptual selection, or inherited consumer-layer bands. That historical confirmation used all 28 decoder layers for MuSiQue and layers 12, 20, 24, and 27 for 2Wiki. A later transport audit, reported below, identified a direct-query position mismatch in that path. The materialization comparisons remain paired within the historical transport, but their absolute memory-versus-no-memory values are not used to choose a deployment layer profile. Contextual K/V is encoded in 256-token blocks, oracle search parents contain 32 tokens, the direct prompt budget is 128 tokens, and the native operation limit is 4096. Validation uses 16 identities per dataset and selects the smallest radius within .05 nats/token of whole-parent likelihood, otherwise the best measured radius. The identity-disjoint held-out cohort contains 32 examples per dataset. Policies, thresholds, and cohort order are frozen before held-out evaluation.

8 Pretrained Oracle Frontier Results

8.1 Validation selects exact evidence

The validation cohort contains 16 examples per dataset. Exact evidence uses 476.3 rather than 547.4 unique K/V tokens on MuSiQue and improves mean gold-token log probability from −7.106 to −7.006. On 2Wiki it uses 79.6 rather than 152.0 tokens and improves likelihood from −4.344 to −4.130. Exact evidence is the smallest candidate within .05 nats/token of the whole-parent control on both datasets, so the predeclared rule selects radius zero. MuSiQue radius 4 has higher raw validation likelihood (−6.786), but it is not the minimum sufficient disclosure under that rule. Held-out policies are not changed after observing this difference.

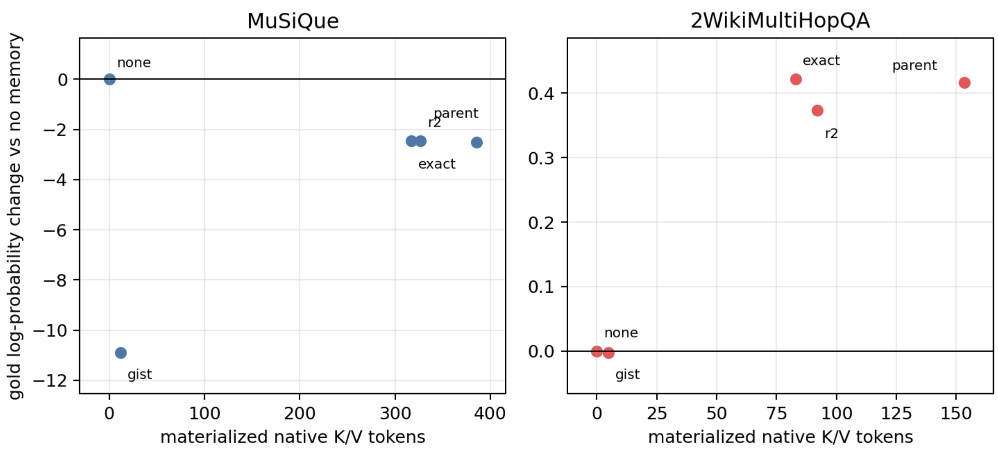

Figure 6: Held-out pretrained K/V–quality frontier. Labels identify the fixed conditions; quality is gold-token log-probability change relative to no memory.

8.2 Exact disclosure preserves the whole-parent result

Table 8 gives the 32-example-per-dataset held-out result. Exact evidence reduces active K/V by 20.9% on MuSiQue and 44.4% on 2Wiki. Its likelihood difference from whole parents is +0.058 nats/token on MuSiQue and +0.005 on 2Wiki. Paired example bootstrap intervals are $[-0.446, 0.518]$ and $[-0.245, 0.202]$, respectively. The data therefore support preservation at a smaller physical point; they do not establish that exact evidence improves likelihood.

Table 8: Pretrained held-out oracle disclosure, 32 examples per dataset. LP is mean gold-token log probability. Parent-relative LP intervals are paired 95% bootstrap intervals.

Dataset	Policy	K/V	Reduction	Coverage	LP	ΔLP [95% CI]	F1
MuSiQue	no memory	0.0	–	.000	−4.398	–	.025
MuSiQue	whole parent	385.4	0.0%	1.000	−6.917	reference	.017
MuSiQue	evidence only	316.8	20.9%	1.000	−6.859	+.058 [−.446, .518]	.024
MuSiQue	radius 2	326.8	17.9%	1.000	−6.851	+.066 [−.529, .511]	.020
2Wiki	no memory	0.0	–	.000	−4.868	–	.267
2Wiki	whole parent	153.3	0.0%	1.000	−4.451	reference	.220
2Wiki	evidence only	83.0	44.4%	1.000	−4.446	+.005 [−.245, .202]	.220
2Wiki	radius 2	92.1	38.1%	1.000	−4.494	−.044 [−.294, .146]	.262

8.3 Attention dilution

The smaller disclosure concentrates attention in the pretrained model as it did in the controlled one. Evidence attention rises from .556 to .689 on MuSiQue and from .171 to .208 on 2Wiki. Radius 2 lowers those values to .664 and .202 while adding 10.0 and 9.1 K/V tokens, respectively. The effect is mechanistic rather than merely a compression statistic: non-evidence states occupy softmax mass without improving the held-out likelihood estimate.

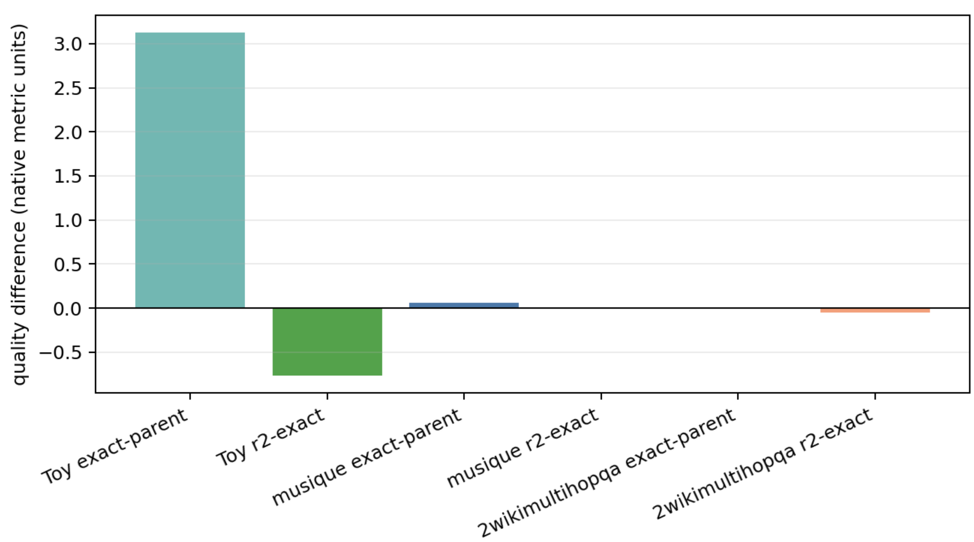

Figure 7: Controlled predictions and pretrained outcomes. Bars use each experiment's native quality metric, so magnitudes are not compared across model classes; signs test transfer.

8.4 Density and coverage remain separate

Exact and radius-2 conditions in the confirmation both retain full annotation coverage. Their density differs: exact evidence is 1.0 by construction, while radius 2 is .962 on MuSiQue and .896 on 2Wiki. Figure 8 keeps the two axes visible. A compact payload can be dense yet incomplete, or complete yet carry avoidable context.

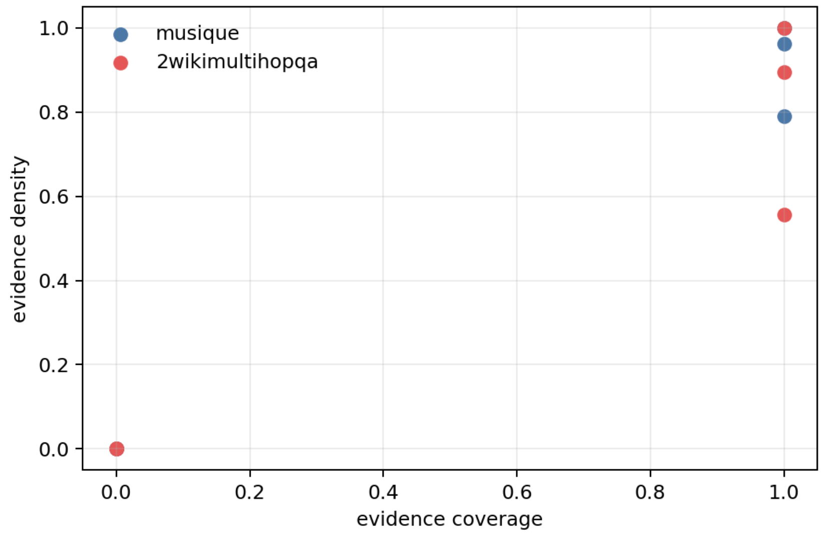

Figure 8: Evidence density and coverage for the pretrained held-out policies. Neither statistic substitutes for the other or for output quality.

The preserved eight-example pilot provides the corresponding failure control. A 128-token MuSiQue payload was .975 dense but covered only .533 of annotated evidence and lost .522 nats/token relative to exact evidence. The same cap covered .966 on 2Wiki and nearly matched whole-parent likelihood. This earlier result is not pooled with the 32-example confirmation, but it explains why "128 tokens are enough" is not a valid materialization rule.

8.5 Gists are routing state, not sufficient detail

Gist-only disclosure reduces K/V by 96.9%, but likelihood is 8.378 nats/token below whole parents on MuSiQue and .419 below on 2Wiki. The controlled T11 condition adds a gist to exact evidence and slightly lowers margin and accuracy. Mean native K/V therefore remains a routing/index control, not a substitute for token detail in either experiment.

8.6 A dataset-specific MuSiQue mismatch

No-memory likelihood is -4.398 on held-out MuSiQue, 2.519 nats/token above whole-parent memory. Exact evidence recovers .058 relative to the parent but remains 2.461 below no memory. By contrast, 2Wiki whole-parent and exact memory improve over no memory by .417 and .422. The materializer is not the source of the MuSiQue gap: exact disclosure preserves the parent result while using less K/V. Nor does MuSiQue establish a universal frozen-consumer limitation. The controlled oracle intervention raises accuracy from .080 to .720 without changing weights, and frozen Qwen consumes 2Wiki memory profitably. MuSiQue instead exhibits a dataset- and interface-specific memory-consumption mismatch that belongs to the consumer/adaptation question.

9 Corrected Transport and Layer Reconciliation

9.1 The replay audit

Native K/V is meaningful only together with its positional frame. The historical pretrained runner retained source-relative post-RoPE keys but restarted the direct query at position zero. The corrected path assigns the query and answer positions immediately after the logical source extent. It also checks that physical GQA/MQA K/V heads are not expanded, interval unions contain no duplicate source positions, source boundaries are clipped, direct and memory logits enter one softmax, and reference detail persists through prefill and decode. These are transport invariants, not tuning choices.

A small paired diagnostic with two HotpotQA and two QASPER identities demonstrates materiality. For all-layer replay, correction changes the mean gold-answer log-probability delta from $+.714$ to $+3.751$ nats on HotpotQA and from -9.775 to -2.971 on QASPER. Last-eight replay changes from $+2.489$ to $+2.089$ and from $+.614$ to $+.277$. Four identities are too few for a quality claim; they are sufficient to reject the old positional path as a basis for layer selection. The public audit records both modes and every checked invariant.

9.2 Four layer roles

The phrase "PRA layer" hid four independently selectable sets. We use

$$\begin{aligned} L_{\text{addr}} &: \text{ layers whose compact states may address memory,} && (11)\\ L_{\text{detail}} &: \text{ layers for which full native source K/V is retained,} && (12)\\ L_{\text{route}} &: \text{ layers that execute routing,} && (13)\\ L_{\text{consume}} &: \text{ layers that join selected native K/V to direct attention.} && (14) \end{aligned}$$

The reconciliation varies L_{consume} while fixing oracle identity, corrected transport, and detail availability. This paper therefore does not imply that every consumer must also route, or that every address layer must retain token-level K/V. A product runtime may store compact address state broadly while retaining detail only where a calibrated consumer can use it.

9.3 Late contiguous bands dominate scattered placement

We evaluate 32 held-out identities, 16 each from MuSiQue and 2Wiki, with the same frozen Qwen3-0.6B model. Whole-parent oracle memory is crossed with all layers, contiguous suffixes of 24, 20, 16, 14, 12, 8, 4, and 1 layers, and matched early, middle, evenly spaced, and late sparse profiles. Gold-answer likelihood is teacher-forced; intervals are paired across profiles.

Table 9: Corrected whole-parent replay. ΔLP is mean gold-answer log-probability relative to no memory; brackets are paired example-bootstrap 95% intervals. States count materialized token–layer K/V states, before the key/value and head-dimension factors.

Dataset	Consumer profile	Layers	ΔLP [95% CI]	States
MuSiQue	all	28	$+.957$ $[-1.403, 2.653]$	13,617
MuSiQue	last 20	20	$+1.946$ $[1.108, 2.873]$	9,726
MuSiQue	last 12	12	$+1.400$ $[.883, 1.988]$	5,836
MuSiQue	last 8	8	$+1.190$ $[.705, 1.812]$	3,891
MuSiQue	last 1	1	$+.005$ $[-.028, .046]$	486
2Wiki	all	28	-2.290 $[-4.822, .027]$	4,144
2Wiki	last 20	20	$+1.276$ $[.610, 2.000]$	2,960
2Wiki	last 12	12	$+1.067$ $[.405, 1.787]$	1,776
2Wiki	last 8	8	$+.949$ $[.288, 1.662]$	1,184
2Wiki	last 1	1	$+.013$ $[-.045, .067]$	148

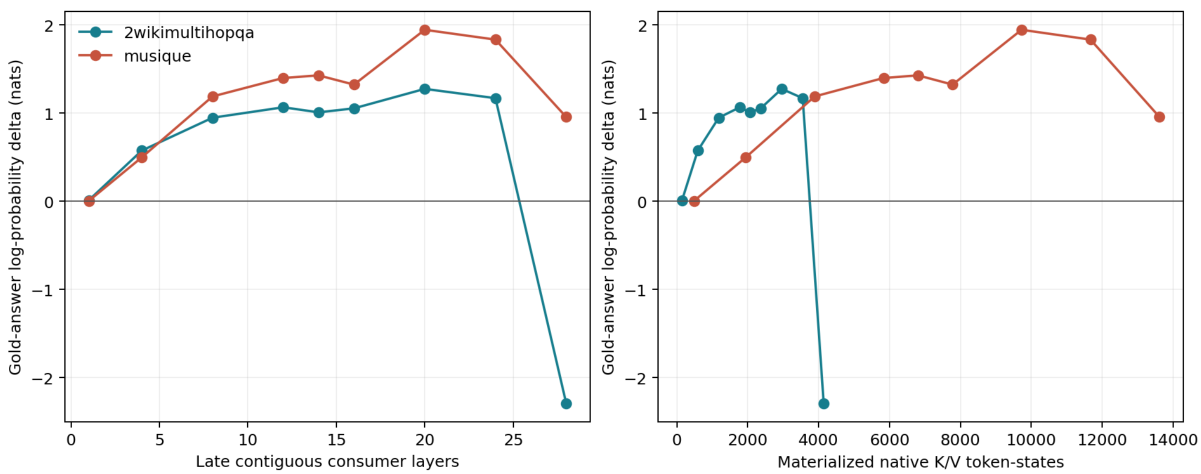

Figure 9: Corrected late-suffix quality versus consumer depth and materialized layer-token cost. The relationship is not monotone: last-20 is best on both workloads, while all-layer replay adds interference.

Placement, not count alone, explains the result. Early-four consumers lower ΔLP by 9.702 nats on MuSiQue and 8.954 on 2Wiki; even-four lowers it by 3.573 and 2.463. Late-four remains positive at .500 and .576. Even-eight is also harmful, whereas the equally sized last-eight band is positive. Thus a request to "use eight PRA layers" is underspecified.

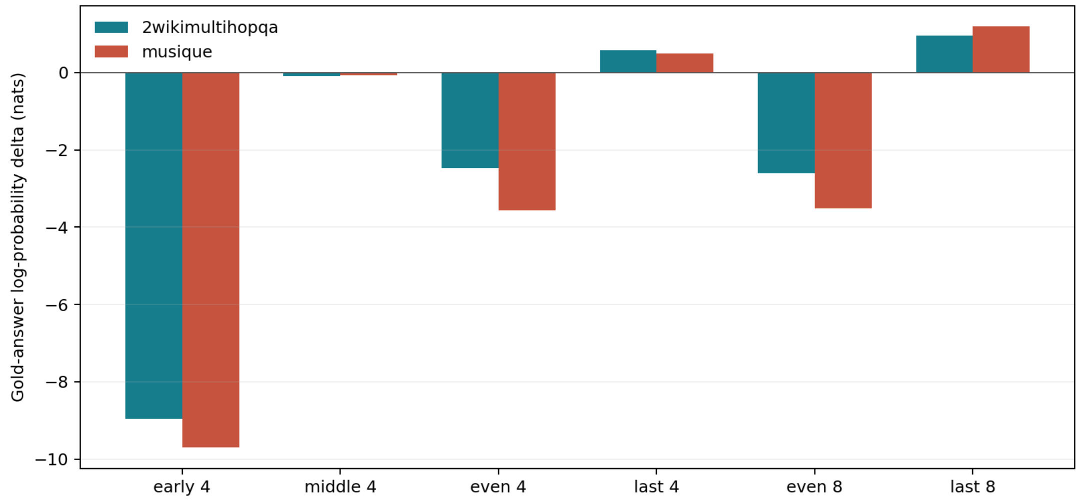

Figure 10: Matched-count sparse and contiguous profiles under corrected transport. Scattered consumption is not an economical substitute for a coherent late computation band.

9.4 Layer placement and materialization width interact

Exact cores, radius-two windows, selected records, and whole parents are crossed with all, last-14, last-eight, and even-eight consumers. Late-14 and late-eight are stable across geometry: their maximum geometry spread is .050 nats on 2Wiki and .035 on MuSiQue. All-layer consumption is not stable. On 2Wiki its ΔLP ranges from -6.36 for exact cores to -2.29 for whole parents; on MuSiQue it ranges from $+.96$ to $+1.33$. Even-eight remains harmful across widths. Materialization cannot repair a poor consumer topology, and a geometry conclusion measured at one topology need not transfer to another.

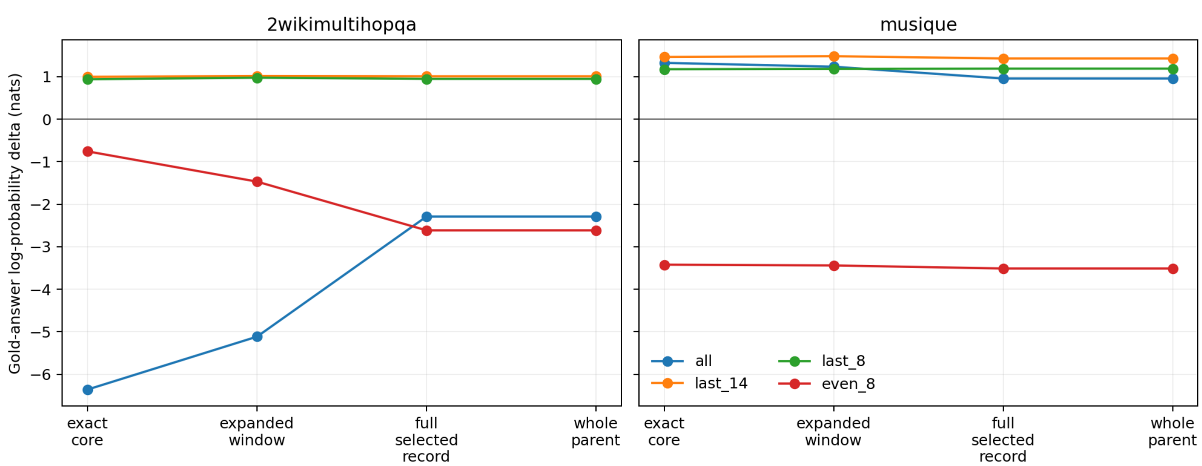

Figure 11: Materialization geometry crossed with consumer placement. The late contiguous bands are robust; all-layer and evenly spaced consumption expose workload-specific interference.

9.5 Profile objectives and handoff

No universal suffix is inferred from 32 identities and one model. The calibration nevertheless supports named objectives. *Quality-max* is last-20 on both workloads. The shared *balanced* profile is last-eight; a dataset-specific rule chooses last-12 for MuSiQue and last-eight for 2Wiki when retaining at least 70% of the best positive mean at minimum state cost. *Economy* is last-one only as a near-zero-gain systems probe, not a quality recommendation. *Reference-correctness* remains all eligible layers when no sparse profile has been validated, but it must not be reported as quality-optimal. The runtime profile registry and cross-model calibration belong to Paper 4.5; this paper supplies the causal placement evidence and physical accounting.

10 Frozen Selection by Materialization

After the oracle confirmation, we reuse one-shot, graph-sparse, graph-balanced, and graph-high Paper-2.5 selected spans. Oracle annotations are unavailable to both selection and interval construction. For each selector we compare the selected 32-token parents, the selector's own atomic logical spans, a 128-token equal allocation, and native gist plus atomic detail.

Table 10: Representative selector/materializer pairs, four held-out examples per dataset. Δ LP and ΔF1 are relative to the same selector with whole-parent disclosure.

Dataset	Selector / disclosure	Parents	K/V	Reduction	ΔLP	ΔF1
MuSiQue	graph sparse / local	13.75	216.2	49.4%	−0.153	.000
MuSiQue	graph sparse / budget 128	13.75	128.0	69.8%	−6.690	.000
MuSiQue	graph balanced / budget 128	11.75	128.0	91.4%	−8.273	.000
2Wiki	graph sparse / local	14.00	221.5	38.2%	+0.069	+.250
2Wiki	graph sparse / budget 128	14.00	128.0	63.3%	+0.223	+.250
2Wiki	graph balanced / budget 128	6.00	128.0	81.8%	+0.280	+.250
2Wiki	graph high / budget 128	9.50	128.0	87.8%	+0.427	+.250

Graph-sparse selection supplies the cleanest direct test because its selected atomic spans are finer than their storage parents. Local materialization nearly halves MuSiQue payload with a modest likelihood loss and reduces 2Wiki payload while improving both likelihood and F1. For one-shot, graph-balanced, and graph-high conditions, many selected spans already cover complete 32-token parents or merge into broad contiguous regions, so local materialization is identical to whole-parent disclosure. Multi-scale discovery does not guarantee a finer physical payload if its final selected intervals are coarse.

The hard budget again exposes a dataset interaction. Every reported 2Wiki selector reaches F1 .75 under 128 tokens, compared with .50 for its whole-parent counterpart. MuSiQue remains at the zero-F1 floor and suffers large likelihood losses. Smart materialization can bound broad conceptual discovery on 2Wiki, but a common global budget does not yet handle MuSiQue's distributed multi-region evidence.

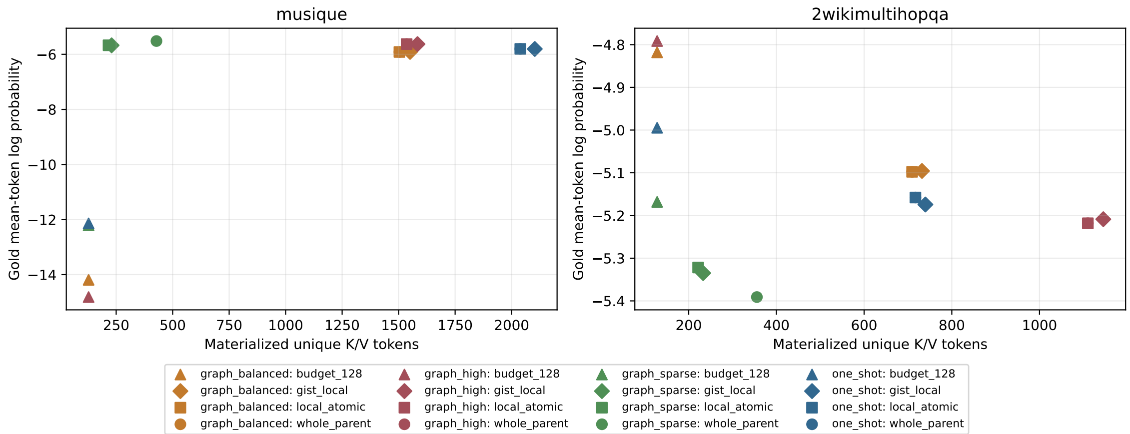

Figure 12: Frozen selection-by-materialization factorial. The same conceptual selection can occupy different physical K/V points; broad selected spans sometimes leave no local compression room.

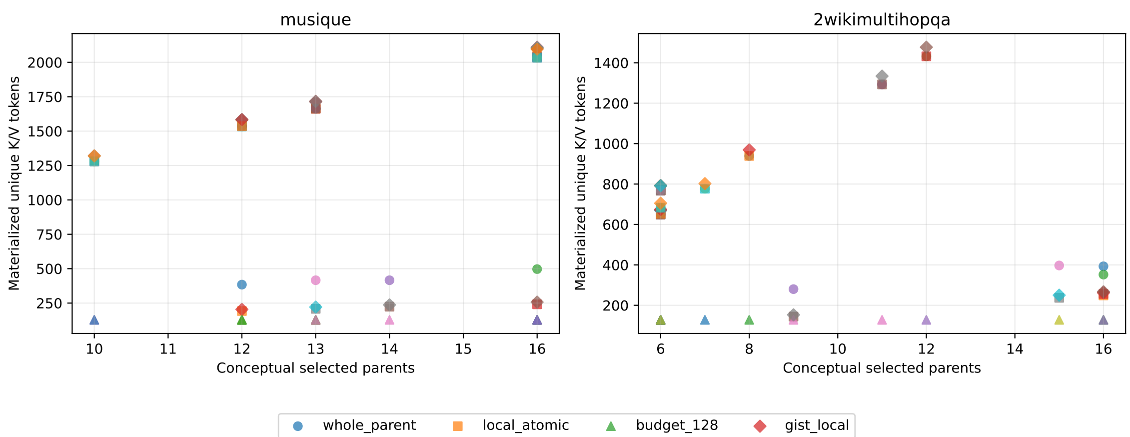

Figure 13: Conceptual parent count versus physical K/V. Fixed budgets decouple the axes, but quality depends on whether the budget preserves all required evidence regions.

11 Systems Accounting

Reference K/V remains on CPU in native KV-head form. The materializer slices CPU tensors before transfer. Exact disclosure reduces mean active layer-token states from 10,791 to 8,872 on MuSiQue and from 613 to 332 on 2Wiki. The logical path transfers 34.65 and 1.30 MiB per example, respectively. The original whole-parent runner exposed bytes and transfer timing through shared core metric names that the Paper-3 exporter did not collect; this naming mismatch is fixed in the released runner, but missing historical measurements are not reconstructed. Unique K/V and layer-token state comparisons remain directly observed for both paths.

The preserved pilot also records that summed cross-shard interval counts reach 63 per MuSiQue example across 28 active layers and 7 per 2Wiki example across four active layers. Thus semantic evidence commonly crosses storage boundaries; clipping at the selected parent would change the experiment.

Deduplication also has observable effect. Held-out 2Wiki evidence requests average 65.1 tokens before union and materialize 62.5, a 4.0% reduction. Validation radius 64 requests 338.8 and materializes 310.3, an 8.4% reduction. MuSiQue's selected examples contain little interval overlap, so deduplication saves little there. Deduplication is semantically required even when it does not dominate runtime.

TTFT does not monotonically follow selected K/V in synchronized single-example execution. In the larger confirmation, evidence only is .059 seconds slower than whole parents on MuSiQue and .027 seconds slower on 2Wiki. Python interval resolution, many small slices, eager attention, Windows WDDM, and single-example noise dominate the token reduction. The result is a memory frontier, not a speed frontier. Fused interval indexing and batched gathers belong to a systems follow-up.

12 Implementation and Reproducibility

The implementation follows the method's ownership boundaries. Logical planning begins at `src/pra_torch/materialization.py:35`, where a half-open interval owns a domain URI and source-relative endpoints. Line 81 attaches the same provenance to native K/V shards. Line 225 sorts and unions overlap only within a domain, and line 312 allocates integer region budgets while respecting capacity and minimum-core constraints.

Physical resolution is separate. Lines 378–401 advance a cursor through every requested interval and reject uncovered ranges. Line 403 slices the covering shards, concatenates them in logical order, preserves tensors shaped $[1, H_{kv}, T, D_h]$, and records tokens, shards, bytes, and transfer time. The shared execution core invokes this path at `src/pra_torch/core.py:249`; line 359 constructs native gist controls, while lines 458–507 dispatch whole, gist, and logical-interval modes behind the same selected identities.

Mechanistic capture is intentionally compact. The joint native attention routine at `src/pra_torch/memory_batching.py:229` optionally retains only final-query memory weights, not full attention matrices. Lines 323–351 package those weights by batch and head, and `src/pra_torch/model.py:703` places them in the progressive trace. This is sufficient to partition evidence, context, distractor, and direct attention without changing logits.

HF reference construction begins at `src/pra_torch/hf/injection.py:308`. Lines 339–446 encode model-safe blocks, preserve global source positions, and record encoding provenance. The pretrained runner checkpoints every condition and releases selected-hit ownership after every example, preventing two complete CPU/GPU-backed reference caches from overlapping on a 4 GB GPU.

The oracle harness, selector factorial, and analysis are in `experiments/paper3_kv_materialization`. Canonical JSON/CSV rows and figures are under `docs/papers/shared/results/paper3_kv_materialization`. Each experiment writes a JSONL checkpoint after every condition. Dataset annotations enter oracle interval planning but are explicitly unavailable to the selector-factorial routers and materializers.

13 Related Work

Sparse-attention architectures reduce the token pairs processed inside a model through fixed or learned patterns [1]. Recurrent and compressed-context approaches retain history through cache policies or summary state; StreamingLLM studies stable attention sinks for streaming caches [8], while Infini-attention introduces compressive memory inside attention [3]. Memorizing Transformers retrieve approximate nearest-neighbor K/V from prior examples [7], and landmark attention separates block-level addressing from token-level access [2]. These systems differ in training and storage contracts, but they motivate separating compact addressing from detailed state.

PRA's materialization question is narrower. The selected payload is layer-native contextual K/V, the host backbone is frozen, and the experiment begins with oracle evidence so retrieval recall cannot confound disclosure. The contribution is not a new end-to-end long-context benchmark leader. It is a controlled physical interface and evidence about how output changes along its K/V frontier.

14 Limitations

The controlled study uses 25 checkpoints but only eight held-out generated identities per checkpoint; examples are repeated across seeds and receptive fields, so 200 checkpoint–example pairs are not treated as 200 independent tasks. The pretrained confirmation has 32 paired identities per dataset and reports example-bootstrap intervals, but those intervals remain broad. Greedy F1 and normalized accuracy are coarse, especially on MuSiQue where output scores remain near the floor. A validated blinded LLM judge has not been run for Paper 3.

Only Qwen3-0.6B is evaluated. Materialization behavior may differ with model scale, family, fine-tuning, quantization, and attention kernel. The corrected layer calibration contains 16 identities per dataset and reuses the same held-out cohort across profiles. Its bootstrap intervals quantify example variation, not model-family or training-seed uncertainty. Profiles must therefore be recalibrated rather than transferred by layer number.

The evidence oracle assumes dataset annotations identify useful source spans. Eighteen MuSiQue and 24 2Wiki held-out identities inherit frozen Paper-2.5 token geometry; the remaining 14 and 8 map the same dataset-truth character spans directly through tokenizer offsets. Tests enforce partition isolation and token-domain alignment, but this extension is not an independently curated benchmark split. An annotation also does not prove every paragraph token is causally necessary, which is precisely why annotation evidence and computational sufficiency are separated.

The direct full-context control OOMed on the available 4 GB GPU under dense eager attention. Latency uses a single Windows GPU without loaded concurrency, and Python gather overhead remains. The paper therefore makes no production throughput, cost, or asymptotic speed claim.

15 Discussion and Handoffs

The experiments tell a more conditional story than a single materialization curve. Exact controlled cores dominate broader disclosure, and exact Qwen evidence preserves its paired whole-parent control while reducing physical state. Corrected replay then shows that consumer placement can dominate that geometry effect: coherent late bands help both workloads, while early or scattered consumers hurt them. Surrounding context and additional consumer layers are therefore resources to calibrate, not automatic support.

The result also clarifies what this paper does not solve. Exact MuSiQue disclosure preserves its parent control but cannot make that control competitive with no memory. Changing the consumer's weights or PRA layer interface may address that mismatch, but doing so here would destroy the frozen causal comparison. The falsifiable Paper-4 hypothesis is that PRA-aware training can make selected memory states more portable and independently consumable. Paper 3 measures the present dependency; Paper 4 must change weights and test whether it moves.

A separate systems follow-up receives logical intervals, exact cross-shard gather, layer-token accounting, and the observed Python transport overhead. It should own fused indexing, batched gather, concurrency, and serving measurements. Unequal-region tasks are also needed before choosing among equal, proportional, and minimum-core budget allocation, because those policies are identical on the balanced controlled cores.

16 Conclusion

Finding memory, loading memory, and choosing where the model consumes it are different operations. Under oracle selection, exact evidence produces a content-causal margin gain in controlled models and preserves its paired whole-parent Qwen control at lower physical cost. A corrected positional audit shows that coherent late consumer bands outperform both all-layer and scattered replay on MuSiQue and 2Wiki. Fixed budgets remain valid only when they preserve every required region; high density can hide low coverage, and more active layers can amplify interference. Computational sufficiency is therefore jointly indexed by materialized content, transport geometry, and consumer placement.

A Test Matrix

Focused tests cover intervals inside one shard, one-boundary and multi-boundary gathers, overlapping storage shards, overlapping requested intervals, exact token order, resource start/end clipping, explicit-resource rejection, `#__head`-to-prompt continuity, GQA shape preservation, CPU-to-GPU transfer bytes, fixed budgets, multi-evidence allocation, radius zero, whole-parent equivalence, logical-core metrics, matched wrong-memory budgets, attention-mass conservation, margin computation, portability, region allocation, validation-only radius selection, deterministic cohort extension, annotation-to-token geometry, shared transport metric names, and native gist controls. Existing PRA, Paper-2, and Paper-2.5 regressions run in the full project suite.

B Canonical Row Schema

Every result row records dataset and identity, selection and materialization policy, radii and K/V budget, conceptual parents, intervals, requested and deduplicated tokens, unique tokens, per-layer token states and bytes, evidence/non-evidence tokens, density and coverage, layer band, output and teacher-forced metrics, attention mass, interval/gather/H2D timing, TTFT, TPOT, total generation, throughput, and peak CUDA allocated/reserved bytes. Oracle use is an explicit boolean.

C Negative Result Ledger

Observation	Consequence
Native gist-only collapses quality	Gists remain routing/index state, not a default detail payload.
Broader toy radii lower quality	Nearby native states are not automatically useful support.
Contextual value states change under isolation	Representation dependence exists, but its correlation with disclosure quality is near zero here.
MuSiQue budget 128 covers 53.3%	Density must always be paired with evidence coverage.
Frozen MuSiQue replay loses to no memory	Correct materialization does not solve memory-use calibration.
Broad selected spans do not shrink locally	Finer physical disclosure requires finer selected intervals or a within-parent policy.
Python gather does not improve TTFT	Optimize indexed/fused gather before speed or cost claims.
Balanced cores do not distinguish allocation rules	Test unequal region sizes before choosing equal, proportional, or minimum-core allocation.
Direct full context OOMs	No dense full-context ceiling is reported on the 4 GB device.
No blinded judge yet	Behavioral equivalence is limited to deterministic and teacher-forced metrics.

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